

MATCHMINDAI - PROJECT PRESENTATION DECK

Real-Time Sports Intelligence & 10,000-Run Monte Carlo Match Prediction Engine

Live Web Application URL: matchmindai.streamlit.app		GitHub Source Repository: github.com/Soham-ss/MatchMindAI	
Core Technology Stack: Python 3.14, Streamlit, Scikit-Learn, Plotly		Predictive Engine: 10,000 Monte Carlo Runs + ML RandomForest	
10,000 Match Runs Simulated Per Ball	18 IPL Seasons Mapped 2008–2026	3 AI Sub-Agents Parallel Execution	100% Free Cloud Host Streamlit Community

■ Slide 2: The Problems We Solved (Why We Built MatchMindAi)

■ Problem 1: Outdated Static Data Only	Traditional prediction websites only look at static past averages from years ago. They completely ignore today's live pitch reports, weather changes, player injuries, or stadium dew factors.
■ Problem 2: Messy Raw AI Data Dumps	General AI chatbots return long, messy text links (e.g. Source [1]...) instead of simple, crisp 1-line direct answers.
■ Problem 3: Lack of Match Simulation Engine	Cricket fans and fantasy players cannot simulate thousands of ball-by-ball match scenarios to view expected score distribution ranges.
■ Problem 4: Complex Technical Jargon	Existing sports analytics tools force users to navigate complicated tables and spreadsheets instead of offering an intuitive, conversational AI interface.

Before MatchMindAi: Static tables, outdated historical records, raw URL dumps, no live weather or pitch awareness.

With MatchMindAi: 10,000-run ball-by-ball simulations, real-time Google News scrapers, and instant 1-line factual responses.

■ Slide 3: Our Solution (How MatchMindAi Fixes Everything)

■ 10,000 Monte Carlo Runs	Simulates 10,000 ball-by-ball matches per click based on team strength ratings, venue pitch profiles, and stadium dew factors.
■ 1-Line Factual Answers	Instantly answers player team queries, Orange Cap winners, Purple Cap winners, and Champions for any year (2008–2026).
■ Live Web Intelligence	Scrapes live Google News RSS feeds to incorporate today's pitch, weather, and team squad updates.
■ High-Energy Cyber UI	Dark cyberpunk theme, live stadium video header, cursor depth particle radar, and sticky bottom Chat Box.

■ Why MatchMindAi Wins: By blending **55% Monte Carlo stochastic math** with **45% Machine Learning pattern recognition** and live web scraping, MatchMindAi gives sports fans the highest level of predictive accuracy.

Slide 4: Main Platform Features & Highlights

Recognizes user name introductions personally (e.g. 'Hi my name is Soham' → 'Hello Soham! ■').	■ 18-Year IPL Database	Map Dav
Detects invalid pre-2008 queries ('who won in 2005') explaining IPL started in 2008 with Shaun Marsh.	■ 1-Click Match Predictor	Pick histo

- Supported Chat Prompts:
- 'who wins 2018 ipl' → ■ Winner: Chennai Super Kings (CSK)
 - 'who win orange cap in 2017' → ■ Orange Cap: David Warner (641 runs)
 - 'who won in 2005' → ■■ Note: IPL was founded in 2008.

■ ■ Slide 5: System Architecture & Workflow (How Data Flows)

Step 1: User Input	User inputs a question or selects 2 teams & stadium venue.
Step 2: Intent Router	Classifies query into Greeting, Bot Identity, IPL History Lookup, or Match Prediction.
Step 3: Web Scraper Agent	Scrapes live Google News RSS feeds for real-time weather & squad updates.
Step 4: Monte Carlo Agent	Executes 10,000 ball-by-ball stochastic match outcome simulations.
Step 5: ML Agent	Queries pre-trained RandomForest model on historical IPL match features.
Step 6: Blended Ensemble	Combines 55% Monte Carlo + 45% ML to compute final win probabilities and report.

■ Ensemble Win Probability Formula:
Final Win Probability = (0.55 × Monte Carlo Win %) + (0.45 × RandomForest ML Win %)

■ Slide 6: Machine Learning & Simulation Math Explained Simply

■ RandomForest Classifier	Trained on historical IPL datasets using scikit-learn. Encodes teams, venue, toss winner, and toss decision. Pre-compiled weights are saved as .pkl binary files in models/.
■ Monte Carlo Math	Calculates run probabilities per ball based on team rating (CSK 88, MI 87, RCB 85) and venue dew factor (Wankhede 0.12). Runs 10,000 iterations in milliseconds.
■ Score Distributions	Generates interactive Plotly bar charts showing expected score ranges and average totals for both competing teams.

■ **Why Random Forest + Monte Carlo?**

Machine learning alone can be rigid if a team has new players. Monte Carlo math simulates randomness (wickets, sixes, extras). Combining both ensures high accuracy even with squad changes.

■ Slide 7: Technical Challenges Faced & Solved

■ GitHub 100 MB Limit Error	■ Solution: Removed IPL.csv (104 MB) from git history because model weights were already compiled into .pkl binary files in models/.
■ Streamlit Cloud Path Errors	■ Solution: Replaced relative string paths with Python's os.path.abspath to dynamically locate exact folder paths on Linux.
■ Missing Cloud Packages	■ Solution: Added beautifulsoup4 and python-dotenv to requirements.txt and wrapped imports in safe try...except blocks.
■ Phrasing Variations	■ Solution: Expanded regex keywords to catch all tense variations ('who wins', 'wins', 'win', 'won').

■■ **Key Takeaway:** Production deployment requires defensive programming — handling missing files, OS path differences, and non-standard user phrasing gracefully.

■ Slide 8: UI/UX & High-Energy Design System

■ Cyberpunk Dark Aesthetic	Designed with a sleek #030712 background and HSL tailored emerald accents (#10b981, #34d399).
■ Modern Typography	Uses Google Font (Outfit) for maximum legibility across laptops, tablets, and mobile devices.
■ Dynamic Visual FX	Features a looping HD stadium lights video banner and an interactive 4D particle cursor depth radar.
■ Sticky Bottom Chat Input	Pinned Chat Box stays permanently visible at the bottom of the viewport when scrolling long chat histories.

■ User Delight: Interactive CSS/JS particle animations make the application feel alive and responsive whenever the user moves their cursor.

■ Slide 9: Cloud Deployment & Infrastructure

■ ■ Live Hosting	Hosted live on Streamlit Community Cloud with 100% uptime. Accessible globally at matchmindai.streamlit.app .
■ Source Control	Version-controlled on GitHub (github.com/Soham-ss/MatchMindAI).
■ Automated CI/CD	Pushing code updates to GitHub main branch automatically triggers a rebuild on Streamlit Cloud within ~30 seconds.

■ **Zero Hosting Cost:** Built using open-source tools and deployed on Streamlit Cloud's free tier, allowing unlimited public user access.

■ Slide 10: Conclusion & Future Roadmap

■ Conclusion	MatchMindAi successfully combines Artificial Intelligence, Machine Learning, Monte Carlo Simulations, and Real-Time Internet Scraping into a fast, accurate, and visually stunning web platform.
■ Live Score API Integration	Connect live cricket APIs for real-time win probability shifts during active live matches.
■ Fantasy Team Optimizer	Auto-generate optimal Dream11 / My11Circle lineups based on Monte Carlo expected points.
■ Multi-Sport Expansion	Extend simulation models to Football (Premier League, Champions League) and Basketball (NBA).